

Calculate the effective data transfer rate for a dedicated channel, given these characteristics:

- **Data is transmitted in blocks of 48 bytes at 100 Mbps**
- **an 8-bit BCC is used for error detection, and an error is detected, on average, once every 1000 transmitted blocks.**

To begin, I would like to briefly define BCC and why it is considered overhead in this context. The Block Check Character (BCC) is a character appended to the end of each block that is used for error detection. It allows the receiver to detect whether any errors occurred during the transmission process. It is considered overhead because overhead refers to any bits that are transmitted but carry no actual user-defined data. As important as the BCC is for data integrity through transmission, we will not count it in this scenario because we are asked to find the effective transfer rate, which refers to how fast solely useful data moves.

We are told one block is 48 bytes ($48 \times 8 = 384$ bits). We then add the eight additional BCC overhead bits to get a final **bit count of 392**.

We can now calculate the total time to transmit one block (including BCC overhead) by dividing 392 by 1,000,000 (one megabit) to get 0.000392, which is **3.92 microseconds**.

We must now account for retransmission by taking into account that out of every 1000 blocks transmitted, one is retransmitted. If we add the one extra retransmission to our total transmission count and divide by 1000 ($1001/1000$), we get 1.001. 1.001 is the multiplier for the average number of transmissions with account for error. We can now multiply 1,001 by the total time per transmission we already calculated (0.000392) to get **3.9239 microseconds**. This is the average time for transmitting one block.

Now since we are told to calculate the *effective* data transfer rate, we must exclude the 8-bit BCC overhead, as that does not count as effective (or useful) data. To finally calculate our data transfer rate, we divide the total number of useful bits by the average time per block transmission ($384/3.9239$). This comes out to about 97.861, which is roughly **97.86 Mbps**.